INCH-POUND

GGG-R-180F
AMENDMENT 1
10 May 1993

FEDERAL SPECIFICATION

REAMERS, HAND AND MACHINE: ARBORS, SHELL REAMER:
AND PILOTS, ADJUSTABLE HAND REAMER

This amendment forms a part of GGG-R-180F, dated 21 January 1992, and is approved for use by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

PAGE 1

Beneficial comments block, line 4: change "SMCAR-EST-S" TO "SMCAR-EST-P".

Under FSC 3455: Delete "3460" and "5110".

PAGE 2

1.2.1, Type X, Class 1, Style A: Delete "Straight shank" and substitute "Straight fluted".

PAGE 4

2.2, line 10: Add "ASTM E 18 - Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials".

PAGE 5

3.1, line 2: Delete "inspection." and substitute "inspection and tests (see 4.2.1).".

3.3, line 6: Add to paragraph 3.3 "Reamers, arbors, or pilots shall be tested for hidden breaks and quench cracks.".

PAGE 6

3.7, line 4: Delete "than C61 and" and substitute "than C63 and".

PAGE 14

3.14.4, line 1: Delete "(machine). the" and substitute "(machine). The".

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PAGE 24

3.24: Delete 3.24 in its entirety and substitute:

"3.24 Type XIV, class 1 and 2, Brown and Sharpe standard taper socket (hand). The reamer shall be a finishing reamer made from high-speed steel and shall have a straight or left hand spiral fluted taper section as specified (see 6.2). The reamers shank shall be provided with a driving square. The chamfer on the cutting end of the reamer is optional. The cutting section of these reamers shall be properly tapered for reaming Brown and Sharpe standard taper sockets used for tools with Brown and Sharpe standard taper socket shanks. Reamers shall be similar to figure 12 and shall conform to Table XI.

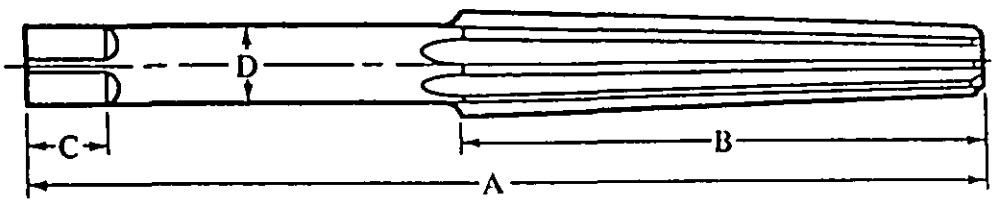


FIGURE 12. Type XIV, class 1 and 2, Brown and Sharpe Taper Socket Reamers - Straight and Left Hand Spiral Flutes, Squared Shank.

TABLE XI. Type XIV, class 1 and 2, Brown and Sharpe Taper Socket Reamers - Straight and Left Hand Spiral Flutes, Squared Shank.

Taper No.	Taper Per Foot	Diameter		Overall Length	Flute Length	Square Length	Shank Dia.	Size of Square	No. of Flutes
		Large End	Small End	A	B	C	D		
1	.50200	0.3176	0.1974	4-3/4	2-7/8	1/4	9/32	0.210	4 to 6
2	.50200	0.3781	0.2474	5-1/8	3-1/8	5/16	11/32	0.255	4 to 6
3	.50200	0.4510	0.3099	5-1/2	3-3/8	3/8	13/32	0.305	4 to 6
4	.50240	0.5017	0.3474	5-7/8	3-11/16	7/16	7/16	0.330	4 to 6
5	.50160	0.6145	0.4474	6-3/8	4	1/2	9/16	0.420	4 to 6
6	.50329	0.6808	0.4974	6-7/8	4-3/8	5/8	5/8	0.470	4 to 6
7	.50147	0.8011	0.5974	7-1/2	4-7/8	3/4	3/4	0.560	6 to 8
8	.50100	0.9770	0.7474	8-1/8	5-1/2	13/16	13/16	0.610	6 to 8
9	.50085	1.1530	0.8974	8-7/8	6-1/8	7/8	1	0.750	6 to 8
10	.51612	1.3376	1.0420	9-3/4	6-7/8	1	1-1/8	0.845	6 to 8

All dimensions are given in inches.

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Tolerances: On overall length (A) and flute length (B), taper numbers 1 to 7, inclusive, ±1/16 inch; taper numbers 8 to 10, inclusive, ±3/32 inch. On square length (C); taper numbers 1 to 9, inclusive, ±1/32 inch; taper number 10, ±1/16 inch. On shank diameter (D), -.0005 to -.002 inch. On size of square, taper numbers 1 to 3, inclusive, -.004 inch; taper numbers 4 to 9, inclusive, -.006 inch; taper number 10, -.008 inch.".

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4.1.1, Delete 4.1.1 in its entirety and substitute:

"4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractors overall inspection system or quality program. The absence of any inspection requirements in the specification does not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirement of the contract.".

4.2.1, Delete 4.2.1 in its entirety and substitute:

"4.2.1 First article inspection. First article inspection shall be performed on one or more reamers, arbors, and pilots when a first article is required (see 3.1 and 6.2). This inspection and test shall include the examination of 4.4 and the test of 4.5 through 4.5.2.18.".

PAGE 27

TABLE XI, Delete Table XI in its entirety.

4.2.2, Delete 4.2.2 in its entirety and substitute:

"4.2.2 Quality conformance inspection. Each sample taken from the inspection lot shall be examined for compliance with 4.4 and 4.5 and subjected to the applicable reaming tests of 4.5 and the packaging inspection of 4.6 (see paragraph 4.3).".

4.3, Delete 4.3 in its entirety and substitute:

"4.3 Sampling. Sampling for quality conformance inspection shall be in accordance with the sampling table XII utilizing the inspection levels specified herein. All samples will be selected randomly from the inspection lot.

Inspection or test	Inspection Level
Product examination (see 4.4)	VI
Hardness test (see 4.5.1)	VI
Performance test (see 4.5.2)	X
Packaging inspection (see 4.6)	VIII

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TABLE XII. Sampling.

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	1/	1/	1/	1/	1/	1/	1/	1/	5	3	2
9 to 15	1/	1/	1/	1/	1/	1/	13	8	5	3	2
16 to 25	1/	1/	1/	1/	1/	20	13	8	5	3	3
26 to 50	1/	1/	1/	1/	32	20	13	8	5	5	5
51 to 90	1/	1/	1/	50	32	20	13	8	7	6	5
91 to 150	1/	1/	125	50	32	20	13	12	11	7	6
151 to 280	1/	1/	125	50	32	20	20	19	13	10	7
281 to 500	1/	315	125	50	48	47	29	21	16	11	9
501 to 1200	1/	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

NOTE:

1/. Indicates one hundred percent inspection. If sample size exceeds lot size perform one hundred percent inspection.

Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.".

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4.5.1, Delete 4.5.1 in its entirety and substitute:

"4.5.1 Hardness test and defective material inspection. Each sample reamer, arbor, or pilot, as applicable, shall be tested for verification of hardness requirement specified in Section 3 (see 3.7, 3.9, 3.12.3, and 3.18.1.5). Testing for hardness shall be in accordance with ASTM E 18. If sufficient area on the cutting edge is not present to take a Rockwell "C" reading, standard files of known hardness may be used. Magnetic particle inspection, X-ray, or other means as approved by the procuring activity shall be employed to inspect the reamers, arbors, or pilots for hidden breaks and quenching cracks.".

4.5.2.1, line 4: Delete "reamer of 1-1/2" and substitute "reamer or 1-1/2".

4.5.2.1, line 5: Delete "material. the" and substitute "material. The".

4.5.2.2, line 4: Delete "desired. a" and substitute "desired. A".

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4.5.2.4, line 4: Delete "desired. the" and substitute "desired. The".

4.5.2.4, line 5: Delete last sentence.

4.5.2.5, Delete 4.5.2.5 in its entirety and substitute:

"4.5.2.5 Type III, classes 4 and 5 (machine reamer) test. The machine reamers shall be tested by reaming a minimum of 100 acceptable holes in 1-1/4 inch thick steel. The test holes shall be prepared 0.012 to 0.015 inch smaller than the size of the hole desired. The reamer shall be operated at a speed in revolutions per minute that gives approximately 80 sfm for any size tested with a feed rate of 1-1/2 percent of the reamer diameter per revolution. Reaming shall produce smooth and round holes. A straight plug gage the size of the finished hole shall be used to measure the size of the hole. Failure of the reamer shall be considered to have occurred when the plug gage will not enter the hole before the minimum number of acceptable holes is reached.".

PAGE 30

4.5.2.13, line 2: Delete "in 1-1;4 inch" and substitute "in 1-1/4 inch".

PAGE 31

4.5.1.17, line 1: Delete "4.5.1.17" and substitute "4.5.2.17".

4.6, line 3: Delete "in paragraph 5" and substitute "in Section 5".

MILITARY INTERESTS:

Custodians
Army - AL
Air Force - 99

Review activities
Army - GL
Air Force - 84

User activity
Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:

FSA-FSS

PREPARING ACTIVITY:
Army - AL

Project No. 5110-0316

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